Year 1 AS Pure Maths

Chapter 12

Differentiation

Practice Booklet

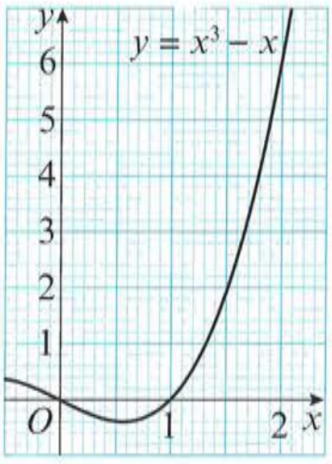
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12.1 Gradients of curves

Practice Book, printed page 93.

1. The diagram shows the curve with equation $y = x^3 - x$.



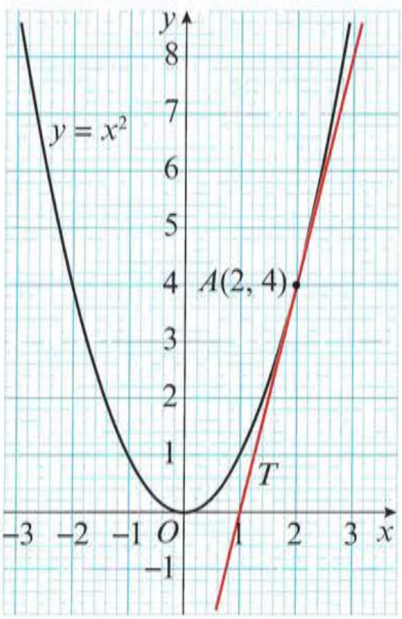
a. Copy and complete this table showing estimates for the gradient of the curve at different values of x.

x-coordinate	0	1	1.5	2
Estimate for gradient of curve				

b. Write a hypothesis about the gradient of the curve at the point where $x = p$.

c. Test your hypothesis by estimating the gradient of the curve at the point $(0.5, -0.375)$.

2. The diagram shows the curve with equation $y = x^2$. The tangent, T, to the curve at the point $A(2, 4)$ is shown.



a. Calculate the gradient of the tangent, T.

b. Calculate the gradient of the chord AP when P has coordinates:

i. $(3, 9)$

ii. $(2.5, 6.25)$

iii. $(2.1, 4.41)$

iv. $(2.01, 4.0401)$

v. $(2 + h, (2 + h)^2)$

12.2 Finding the derivative

Practice Book, printed page 93–94.

1. For the function $f(x) = 12x$, use the definition of the derivative to show that:
   - **a.** $f'(4) = 12$
   - **b.** $f'(-2) = 12$
   - **c.** $f'(0) = 12$

Hint

The derivative is defined by

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Substitute the value given for x into the definition of the derivative. Then use the function to expand and simplify. In each case you should be able to cancel h from each term in the numerator.

2. For the function $f(x) = 2x^2$, use the definition of the derivative to show that:
   - **a.** $f'(3) = 12$
   - **b.** $f'(-1) = -4$
   - **c.** $f'(0) = 0$

3. E/P Prove, from first principles, that the derivative of $8x$ is 8. (3 marks)

4. E/P Prove, from first principles, that the derivative of $3x^2$ is $6x$. (4 marks)

5. E/P Prove, from first principles, that the derivative of x^3 is $3x^2$. (4 marks)

6. E/P The curve C has equation $6 - 3x^2$. The curve C passes through the point Q with x-coordinate 1. Use differentiation from first principles to find the value of the gradient of the tangent to C at Q. (5 marks)

12.3 Differentiating x^n

Practice Book, printed page 94.

1. Find $f'(x)$ given that $f(x)$ equals:
   - **a.** x^5
   - **b.** x^{-6}
   - **c.** $\sqrt{x}$
   - **d.** $\frac{1}{\sqrt[4]{x}}$

Hint

First write each function in the form x^n.

2. Find $\frac{dy}{dx}$ given that y equals:
   - **a.** $5x^4$
   - **b.** $5x^{-2}$
   - **c.** $\frac{3x^2}{12x^5}$
   - **d.** $\frac{9}{\sqrt[3]{x^2}}$

Hint

If $y = ax^n$, then $\frac{dy}{dx} = anx^{n-1}$.

3. Find the gradient of the curve $y = 2x^3$ at the point where:

a. $x = 3$

b. $x = -5$

c. $x = \frac{1}{2}$

d. $x = \frac{3}{4}$

Hint

First differentiate $2x^3$ and then substitute each value of x.

4. E Find the gradient of the curve $y = \frac{5}{x^3}$ at the point where $x = 2$. (2 marks)

5. E Find the gradient of the curve $y = -\frac{8}{\sqrt{x}}$ at the point where $x = 16$. (2 marks)

6. E/P The gradient of the curve $y = 2x^4$ at the point $x = a$ is 27. Find the value of a. (3 marks)

7. E/P A curve C has equation $y^3 - 64x^2 = 0$, $y \geq 0$. Find the gradient of C at the point where $x = 27$. (4 marks)

12.4 Differentiating quadratics

Practice Book, printed page 94–95.

1. Find $f'(x)$ given that $f(x)$ equals:

a. $3x^2 - 7x + 12$

b. $\frac{3}{4}x^2 + \frac{1}{2}x - 1$

c. $12 - 5x^2$

Hint

Differentiate the terms one at a time. If $f(x) = ax^2 + bx + c$, then $f'(x) = 2ax + b$.

2. Find the gradient of the curve with the equation:

a. $y = -4x^2 + 11x - 1$ at the point $(2, 5)$

b. $y = 3x^2 - 20$ at the point $(4, 28)$

c. $y = 18 + \frac{7}{2}x - \frac{1}{4}x^2$ at the point $(8, 30)$.

Hint

Substitute the x-value into the expression for $\frac{dy}{dx}$ to find the gradient at that point.

3. P Find the coordinates of the point on the curve with equation $y = 2x^2 - 3x + 4$ where the gradient is 9.

Hint

Set $\frac{dy}{dx} = 9$ and then solve to find the value of x. Then find the corresponding value of y.

4. E/P Find the gradient of the curve with the equation $y = (2 - x)(2 + x)$ at the point where $x = 5$. (3 marks)

5. E/P The curve with equation $y = 6x^2 - 5x + 3$ has gradient -29 when $x = p$. Find the value of p. (3 marks)

6. E The curve with equation $y = x^2 - 3x + 1$ intersects the line with equation $y = 2x - 5$ at the points P and Q. Find:

a. the coordinates of the points P and Q (3 marks)

b. the gradient of the curve at the points P and Q. (3 marks)

7. E The curve with equation $y = 5x^2 - 6x - 1$ intersects the line with equation $y = 7$ at the points A and B. Find:

a. the coordinates of the points A and B (3 marks)

b. the gradient of the curve at the points A and B. (3 marks)

12.5 Differentiating functions with two or more terms

Practice Book, printed page 95.

1. Differentiate:

a. $5x - 7x^2$

b. $8x^{5/2} - 6x^4$

c. $8x^{5/2} - 4x^2 - 6x^{1/3}$

Hint

Differentiate the terms one at a time.

2. Find $f'(x)$ given that $f(x)$ equals:

a. $\sqrt{x} + \frac{1}{\sqrt{x}}$

b. $\frac{1}{7x^2} + \sqrt[4]{x}$

Hint

First write each term in the form ax^n.

3. Find $\frac{dy}{dx}$ when y equals:

a. $x(2x^2 - 6x + 5)$

b. $(x^2 - 2)(x + \frac{2}{x})$

c. $\frac{1}{\sqrt{x}}(6x^{5/2} - 8x^{3/2})$

Hint

Multiply out the brackets and write each term in the form ax^n.

4. In each case, find the gradient of the curve with equation $y = f(x)$ at the point A:

a. $f(x) = \frac{1}{2}x^4 - 8x^2 + 2$; A is $(-2, -22)$

b. $f(x) = \frac{6}{\sqrt{x}} - 3x - 2x^2$; A is $(1, 1)$

5. E Find the gradient of the curve $f(x) = \frac{4x^2 - 8x}{x^3}$ at the point $P(-2, -4)$. (3 marks)

6. E/P Given that $f(x) = px^2 - 8p\sqrt{x}$ and that $f'(4) = 15$, find the value of p. (4 marks)

7. E/P $h(x) = px^3 + 4px + \frac{q}{x^3}$, where p and q are constants.
Given that $h(-1) = -41$ and $h'(-1) = 67$, find the values of p and q. (5 marks)

8. E/P $f(x) = (1 + 2x)^6$, $g(x) = (1 - 2x)^6$

a. Find the first three terms, in ascending powers of x, of the binomial expansion of $f(x)$, giving each term in its simplest form. (2 marks)

b. If x is small, so that x^2 and higher powers can be ignored, find:

i. $f'(x)$

ii. $g'(x)$

(4 marks)

12.6 Gradients, tangents and normals

Practice Book, printed page 96.

1. Find the equation of the tangent to the curve:

a. $y = x^3 - 4x^2$ at the point $(2, -8)$

b. $y = \frac{4}{x} - 3x^2$ at the point $(-2, -14)$

c. $y = 6x^{3/2} - 4x^{1/2}$ at the point where $x = 4$

d. $y = \frac{12 - x^2}{2x}$ at the point where $x = -3$.

Hint

The tangent to the curve $y = f(x)$ at the point where $x = a$ has equation

$$y - f(a) = f'(a)(x - a)$$

2. Find the equation of the normal to the curve:

a. $y = \frac{1}{2}x^2 + \frac{6}{x}$ at the point $(3, \frac{13}{2})$

b. $y = \frac{4}{x} - 3x^2$ at the point where $x = 4$.

Hint

The normal to the curve $y = f(x)$ at the point where $x = a$ has equation

$$y - f(a) = -\frac{1}{f'(a)}(x - a)$$

3. $g(x) = 10 - 2x - x^2$

a. Find the equation of the tangent to the curve $y = g(x)$ at the point where $x = 1$.

b. Find the equation of the normal to the curve $y = g(x)$ at the point where $x = 1$.

4. E $f(x) = x^2 - 5x + 4$

a. Find the equation of the tangent to the curve $y = f(x)$ at the point where $x = 5$. (3 marks)

b. Find the equation of the normal to the curve $y = f(x)$ at the point where $x = 3$. (3 marks)

The tangent found in part **a** intersects the normal found in part **b** at the point P.

c. Find the coordinates of point P. (4 marks)

5. E $g(x) = x^3 - 4x$

a. Find the equation of the tangent to the curve $y = g(x)$ at the point where $x = -1$. (3 marks)

This tangent meets the x-axis at the point A.

b. Find the coordinates of A. (1 mark)

c. Show that A lies on the curve $y = g(x)$. (1 mark)

6. E/P The point P with x-coordinate 6 lies on the curve with equation $y = \frac{1}{2}x^2 - 8x + 6$.
The normal to the curve at P intersects the curve at the points P and Q.
Find the coordinates of Q. (6 marks)

7. E/P The point $P(4, 2.5)$ lies on the curve C with equation $y = \sqrt{x} + \frac{1}{\sqrt{x}}$.
Giving your answer in the form $ax + by + c = 0$, find:

a. the equation of the tangent to C at P (3 marks)

b. the equation of the normal to C at P. (3 marks)

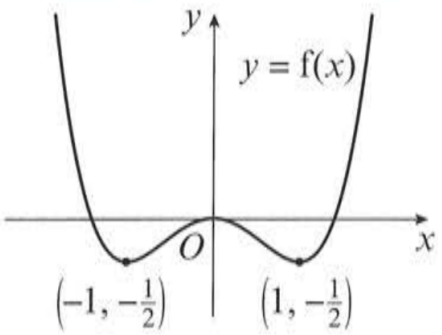
The tangent to C at P intersects the y-axis at A and the normal to C at P intersects the y-axis at B.

c. Find the area of triangle ABP. (4 marks)

12.7 Increasing and decreasing functions

Practice Book, printed page 97.

1. The graph shows the function $y = f(x)$ where $f(x) = \frac{1}{2}x^4 - x^2$.



a. State the coordinates of the three points where the gradient of $y = f(x)$ is zero.

b. State whether the function is increasing, decreasing or neither over the interval:

i. $(-\infty, -1)$

ii. $[-1, 0]$

iii. $[0, 1]$

iv. $(-1, 1)$

v. $[1, \infty)$

vi. $[0, 2]$

Hint

The function $f(x)$ is increasing on $[a, b]$ if $f'(x) \geq 0$ for all values of x such that $a < x < b$.
The function $f(x)$ is decreasing on $[a, b]$ if $f'(x) \leq 0$ for all values of x such that $a < x < b$.
If neither of these conditions holds for all values on a given interval, then the function is neither increasing nor decreasing on that interval.

2. Find the range of values of x for which $f(x)$ is increasing, given that $f(x)$ equals:

a. $5x^2 - 8x + 7$

b. $14 - 9x - 3x^2$

c. $x^3 - 12x + 8$

d. $18x - 2x^3$

3. Find the range of values of x for which $f(x)$ is decreasing, given that $f(x)$ equals:

a. $3x^2 + 7x - 2$

b. $15 - 6x - 5x^2$

c. $-2x^5 - 8x^3$

d. $5x(3 - x)$

4. E/P Show that the function $f(x) = x^5 + 2x^3 + 8x - 18$ is increasing for all real values of x. (3 marks)

5. E/P Show that the function $g(x) = -2x^3 - 18x^2 - 54x + 10$ is decreasing for all real values of x. (3 marks)

6. E/P $h(x) = x^3 + qx$
Find the range of values of q for which $h(x)$ is increasing for all real values of x. (3 marks)

7. E/P A curve has equation $y = 3x^3 + 2x^2 - 5x - 6$.

a. Find $\frac{dy}{dx}$ (2 marks)

b. Hence find the range of values of x for which y is increasing. Write your answer using set notation. (4 marks)

12.8 Second order derivatives

Practice Book, printed page 97–98.

1. Find $f'(x)$ and $f''(x)$ when $f(x)$ equals:

a. $9x^2 - 6x + 7$

b. $-x^3 + 4x^{-1}$

c. $4x^{-1/2} + 8x^{3/2}$

Hint

First differentiate $f(x)$ to obtain $f'(x)$. Then differentiate $f'(x)$ to obtain $f''(x)$.

2. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ when y equals:

a. $\frac{6}{x^3} - \frac{1}{2}x^4 - 9x$

b. $\frac{8}{\sqrt[4]{x}} - \sqrt[6]{x} - 7$

c. $(x^2 - 3)(5 - x^3)$

Hint

Write each term in the form ax^n and then differentiate twice to obtain $\frac{d^2y}{dx^2}$.

3. $h(x) = 4x^4 - \sqrt[3]{x} + \frac{1-x^5}{x^3}$. Find:

a. $h'(x)$

b. $h''(x)$

4. E Given that $f(x) = \frac{1}{x} - 6x - 2x^3$, find $f''(-2)$. (3 marks)

5. E $p(x) = (1 - 4x)^3$

a. Fully expand $p(x)$. (2 marks)

b. Find $p''(x)$. (1 mark)

6. E/P $f(x) = ax^3 + bx^2 - 8x + 6$, where a and b are constants.
Given that $f'(2) = -24$ and $f''(-5) = 100$, find the values of a and b. (4 marks)

7. E/P $g(x) = px^3 + qx^2 + 6x - 1$, where p and q are constants.
Given that $g'(-2) = g'(\frac{5}{2}) = 0$, find:

a. the value of p and the value of q (4 marks)

b. $g''(x)$. (2 marks)

12.9 Stationary points

Practice Book, printed page 98–99.

1. $f(x) = (x-4)(x-8)$

a. Sketch the graph of $y = f(x)$.

b. Explain how the graph in part **a** shows that $y = f(x)$ has a local minimum between $x = 4$ and $x = 8$.

c. Find $f'(x)$.

d. Solve $f'(x) = 0$.

e. State the coordinates of the stationary point.

Hint

The stationary point occurs where $f'(x) = 0$. The shape of the graph of $y = f(x)$ can be used to determine whether this stationary point is a local minimum or a local maximum.

2. Find the coordinates of the minimum point on the curve with equation:

a. $y = x^2 - 6x + 8$

b. $y = x^2 - 3x - 11$

c. $y = 3x^2 + 2x + 1$

Hint

Find $f'(x)$ and then solve $f'(x) = 0$ to obtain the value of x at the stationary point. Then find the corresponding value of $y = f(x)$.

3. Find the coordinates of the maximum point on the curve with equation:

a. $y = 8 - 16x - x^2$

b. $y = 12x - 4x^2$

c. $y = 18 - 30x - 7x^2$

4.

a. Find the coordinates of the stationary points on the curve with equation $y = -4x^3 + 6x^2 + 24x + 3$

b. Find $\frac{d^2y}{dx^2}$

c. Use $\frac{d^2y}{dx^2}$ to determine the nature of the stationary points on the curve of y.

Hint

If $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} > 0$, the stationary point is a local minimum.
If $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} < 0$, the stationary point is a local maximum.

5. $q(x) = -x^3 - 6x^2 - 12x - 6$

a. Find the coordinates of the stationary point on the curve with equation $y = q(x)$.

b. Show that this stationary point is a point of inflection.

c. Sketch the graph of $y = q(x)$.

Hint

If $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} = 0$, the stationary point could be a local minimum, local maximum or a point of inflection.
Look at the gradient either side of the stationary point to determine its nature.

6. E $f(x) = 2x^3 - \frac{1}{2}x^2 - x + 2$

Find the coordinates of the stationary points on the curve with equation $y = f(x)$. (4 marks)

7. E $g(x) = \frac{54}{x} + 4\sqrt{x}$

a. Find the coordinates of the stationary point on the curve $y = g(x)$. (5 marks)

b. Find $g''(x)$. (1 mark)

c. Show that the stationary point from part **a** is a local minimum. (1 mark)

8. E $f(x) = -2x^3 + 4x^2 + 8x - 10$
Find the coordinates of the local maximum on the curve with equation $y = f(x)$. (5 marks)

9. E $h(x) = x^3 - 3x^2 + 3x + 4$

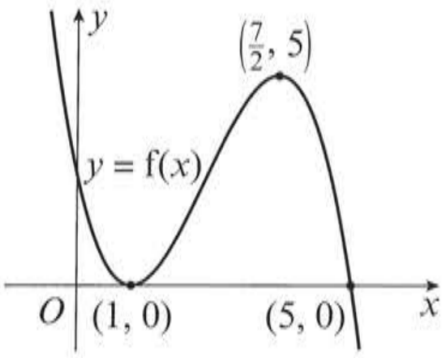
a. Find the coordinates of the stationary point on the curve $y = h(x)$ and determine the nature of this stationary point. (4 marks)

b. Sketch the graph of $y = h(x)$. (2 marks)

12.10 Sketching gradient functions

Practice Book, printed page 99–100.

1. The diagram shows the curve with equation $y = f(x)$.



a. Write down the coordinates of the stationary points on the curve $y = f(x)$.

b. Describe the gradient of $y = f(x)$ between the two stationary points.

c. Describe the gradient of $y = f(x)$ for $x < 1$.

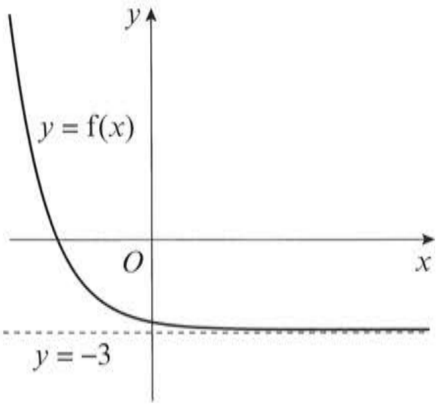
d. Describe the gradient of $y = f(x)$ for $x > \frac{7}{2}$

e. Use your answers to parts **a** to **d** to sketch the graph of $y = f'(x)$.

Hint

At the point where $y = f(x)$ has a stationary point, $y = f'(x)$ will intersect the x-axis.
At the point where $y = f(x)$ has positive gradient, $y = f'(x)$ will be above the x-axis.
At the point where $y = f(x)$ has negative gradient, $y = f'(x)$ will be below the x-axis.

2. The diagram shows the curve with equation $y = f(x)$.

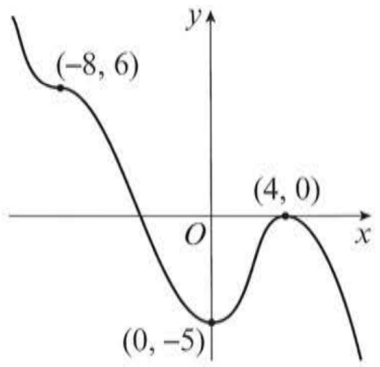


a. Describe the gradient of $y = f(x)$ for all values of x.

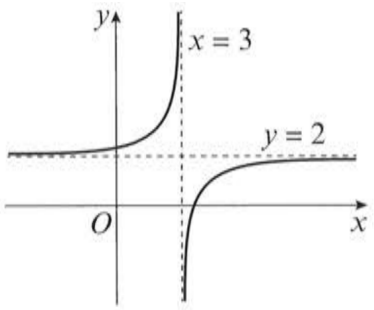
b. Sketch the graph of $y = f'(x)$, giving the equation of any asymptotes.

3. For each graph, sketch the graph of the corresponding gradient function on a separate set of axes. Label the coordinates of any points where the curve cuts or meets the x-axis, and give the equations of any asymptotes.

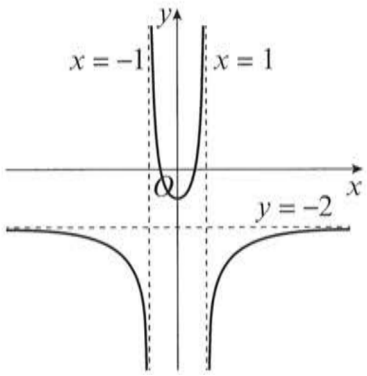
a



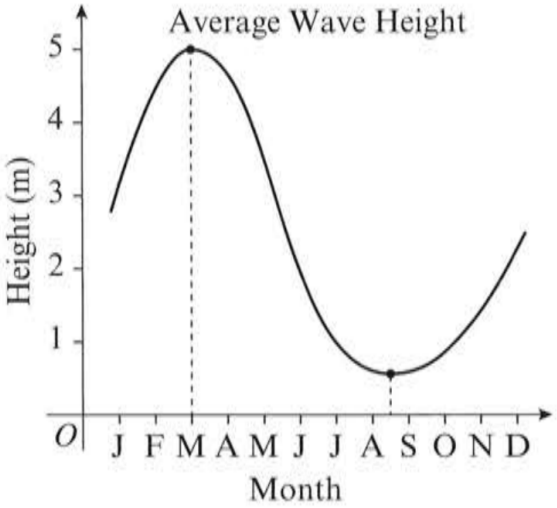
b



c



4. E The graph shows the average wave height at a beach during each month.



On a separate graph, sketch the gradient function for the wave height. Indicate the coordinates of any points where the gradient function crosses the horizontal axis. (3 marks)

12.11 Modelling with differentiation

Practice Book, printed page 101–102.

1. Find $\frac{dA}{dx}$ where $A = 6x^2$.

Hint

Finding $\frac{dA}{dx}$ where $A = 6x^2$ is the same as finding $\frac{dy}{dx}$ where $y = 6x^2$.

2. Given that $V = \frac{20000}{t^2}$, find:

a. $\frac{dV}{dt}$

b. the value of $\frac{dV}{dt}$ when $t = 10$

Hint

Substitute $t = 10$ into your expression for $\frac{dV}{dt}$.

3. When a cyclist reaches the bottom of a large hill, he is travelling with a speed, v, of 8 m s^{-1}. The cyclist then free-wheels along a straight flat road with speed $v = (8 - \sqrt{t})$ m s^{-1}, where t is the time elapsed since he reached the bottom of the hill.
Find:

a. the time it takes for the cyclist to come to a stop

b. $\frac{dv}{dt}$

c. the cyclist's acceleration at time $t = 9$ seconds.

Hint

Acceleration is the rate of change of speed with respect to time: $a = \frac{dv}{dt}$
← Statistics and Mechanics Year 1, Chapter 11

4. The volume, V cm^3, of an expanding sphere of radius r cm is given by $V = \frac{4}{3}\pi r^3$. Find:

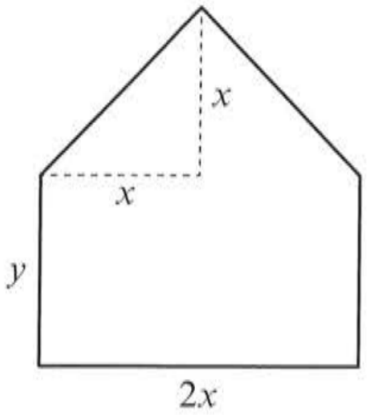
a. the rate of change of the volume with respect to the radius

b. the rate of change of volume with respect to radius when the radius is 4 cm.

Hint

$\frac{dV}{dr}$ is the rate of change of volume with respect to the radius.

5. E/P A fence is 80 m long and is used to enclose an area with five sides, as shown in the diagram. Some of the dimensions of the fence are also shown. The area enclosed by the fence is A m^2.



a. Find an expression for y in terms of x. (3 marks)

b. Prove that $A = 80x - x^2(1 + 2\sqrt{2})$. (3 marks)

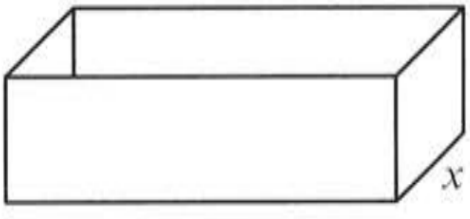
c. Given that x can vary, use calculus to find the exact maximum value of A. (5 marks)

6. E/P A solid cylinder has volume 1024π cm^3 and variable height h and radius r.

a. Show that the surface area of the cylinder is given by $S = \frac{2048\pi}{h} + 64\pi\sqrt{h}$. (3 marks)

b. Hence find the minimum value of the surface area, leaving your answer in terms of π. (5 marks)

7. E/P A rectangular box has a horizontal base but no top.
The cross-section of the box is a square of side length x cm.
The total area of cardboard used to make the box is 608 cm^2.

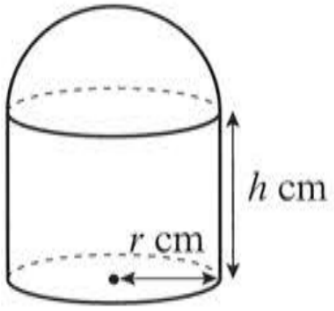


a. Show that the volume, V cm^3, of the box is given by $V = \frac{608x - 2x^3}{3}$ (3 marks)

b. Find the maximum value of V, giving your answer to 1 decimal place. (3 marks)

c. Explain how you know your answer is a maximum. (2 marks)

8. E/P The diagram shows a design for a storage bin. The bin is made from a cylinder of radius r cm and height h cm attached to a hemisphere of radius r cm.



The total surface area of the bin is 800π cm^2, and the volume of the bin is V cm^3.

a. Show that $V = \frac{\pi r(2400-5r^2)}{6}$ (5 marks)

b. Find the maximum volume of the bin. Give your answer correct to 3 significant figures. (5 marks)

Problem set A

Bronze

$h(x) = 14 + 6x - x^2$

a. Find the coordinates of the stationary point on the curve with equation $y = h(x)$. (3 marks)

b. By sketching the graph, or otherwise, determine the nature of the stationary point. (2 marks)

Silver

$g(x) = -x^3 - 3x^2 + 4$.

a. Find the coordinates of the stationary points on the curve with equation $y = g(x)$. (4 marks)

b. Hence, find all the solutions to the equation $g(x) = 0$. (3 marks)

c. By sketching the graph of $y = g(x)$, or otherwise, determine the nature of each stationary point. (2 marks)

Gold

$f(x) = \frac{2}{3}\sqrt{x^3} - 10\sqrt{x} - \frac{8}{\sqrt{x}}$, $x > 0$.

a. Find the coordinates of the stationary points on the curve with equation $y = f(x)$. (5 marks)

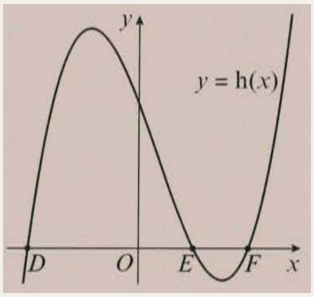
b. By considering $f''(x)$, determine the nature of each stationary point. (4 marks)

Problem set B

Bronze

$h(x) = (x - 2)(x - 1)(x + 2)$

The curve with equation $y = h(x)$ cuts the x-axis at the points D, E and F as shown in the diagram.



a. Write down the coordinates of the points D, E and F. (1 mark)

b. Find $h'(x)$. (2 marks)

c. Find the equation of the tangent to the curve at:

i. E

ii. F

(3 marks)

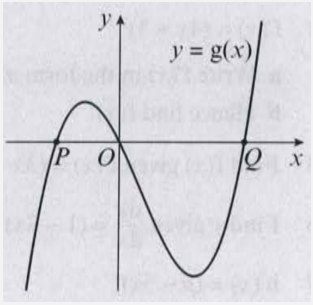
The tangents in part **c** intersect at a point G.

d. Find the coordinates of G. (2 marks)

Silver

$g(x) = x(x^2 - x - 2)$

The curve with equation $y = g(x)$ cuts the x-axis at the origin and at the points P and Q as shown.



a. Find $g'(x)$. (2 marks)

b. Find the equation of the normal to C at:

i. P

ii. Q

(3 marks)

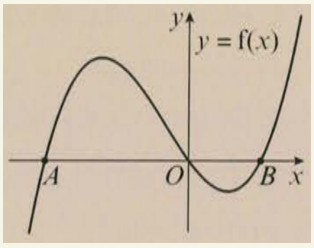
The two normals found in part **b** intersect at a point R.

c. Find the coordinates of R. (2 marks)

Gold

$f(x) = \frac{1}{6}x^3 + \frac{1}{3}x^2 - \frac{4}{3}x$

The curve with equation $y = f(x)$ cuts the x-axis at the origin and at the points A and B as shown. The normal to the curve at A intersects the normal to the curve at B at a point P. Find the area of triangle APB. (8 marks)



Selected exam question bank questions

Question 11

$f(x) = 8x^2 - \frac{80}{3}x^{3/2} + 5x;\ x \geq 0$

Find the x-coordinates of the points on the curve with equation $y = f(x)$ where the gradient is equal to -4. (5 marks)

Question 15

$f(x) = \frac{1}{3}\sqrt{x} + \frac{3}{\sqrt{x}};\ x > 0$

a. Find $f'(x)$. (2 marks)

b. Find the coordinates of the stationary point on the curve $y = f(x)$. (2 marks)

c. By first finding $f''(x)$, determine whether the stationary point found in **b** is a local minimum or a local maximum. (2 marks)

Question 27

$f(x) = ax^3 + bx^2 + cx$

The curve with equation $y = f(x)$ has a gradient of 7 when $x = -1$ and a gradient of 5 when $x = 2$. Additionally, $f''(1) = 2$. Find the values of the constants a, b and c. (6 marks)

Question 50

$f(x) = 4 - 2\sqrt{x} + \frac{1}{2}x^2$

a. Find the coordinates of the stationary point on the curve with equation $y = f(x)$. (3 marks)

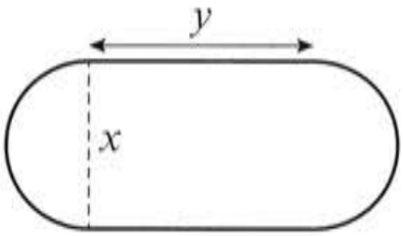
b. Find the equation of the normal to the curve at $P(4, 8)$. (5 marks)

The normal to the curve at P intersects the x-axis at A.

c. Find the area of the triangle OAP. (3 marks)

Question 74

A horse riding facility wishes to create a riding space using 300 m of fencing. The riding space must comprise a rectangular area and two semi-circles, as shown in the diagram.



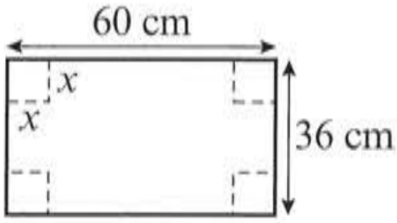
a. Show that the area of the riding space can be expressed by the formula $A = 150x - \frac{\pi}{4}x^2$ (4 marks)

b. Given that x can vary, find the exact maximum area of the riding space. (4 marks)

Question 90

A sheet of cardboard has dimensions 60 cm by 36 cm.

A square of side length x cm is removed from each of the four corners as shown in the diagram. The sheet of cardboard is then folded to make a box.



a. Show that the volume of the box is given by the equation $V = 4x^3 - 192x^2 + 2160x$ (4 marks)

b. Find the value of x for which the volume is a maximum. Leave your answer in the form $a + b\sqrt{19}$, where a and b are constants to be found. (4 marks)

c. Find the maximum volume of the box, giving your answer to one decimal place. (2 marks)

Answers

12.1 Gradients of curves

1.

a. For example:

x-coordinate	0	1	1.5	2
Estimate for gradient of curve	-1	2	5.75	11

b. Gradient $= 3p^2 - 1$

c. -0.25

2.

a. 4

b.

i. 5

ii. 4.5

iii. 4.1

iv. 4.01

v. $4 + h$

12.2 Finding the derivative

1.

a. $f'(4) = \lim_{h\to 0} \dfrac{f(4+h) - f(4)}{h} = \lim_{h\to 0} \dfrac{12(4+h) - 12(4)}{h}$

$= \lim_{h\to 0} \dfrac{12h}{h} = 12$

b. $f'(-2) = \lim_{h\to 0} \dfrac{f(-2+h) - f(-2)}{h}$

$= \lim_{h\to 0} \dfrac{12(-2+h) - 12(-2)}{h} = \lim_{h\to 0} \dfrac{12h}{h} = 12$

c. $f'(0) = \lim_{h\to 0} \dfrac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \dfrac{12(0+h) - 12(0)}{h}$

$= \lim_{h\to 0} \dfrac{12h}{h} = 12$

2.

a. $f'(3) = \lim_{h\to 0} \dfrac{f(3+h) - f(3)}{h} = \lim_{h\to 0} \dfrac{2(3+h)^2 - 2(3)^2}{h}$

$= \lim_{h\to 0} \dfrac{12h + 2h^2}{h} = \lim_{h\to 0}(12 + h) = 12$

b. $f'(-1) = \lim_{h\to 0} \dfrac{f(-1+h) - f(-1)}{h}$

$= \lim_{h\to 0} \dfrac{2(-1+h)^2 - 2(-1)^2}{h}$

$= \lim_{h\to 0} \dfrac{-4h + 2h^2}{h} = \lim_{h\to 0}(-4 + h) = -4$

c. $f'(0) = \lim_{h\to 0} \dfrac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \dfrac{2(0+h)^2 - 2(0)^2}{h}$

$= \lim_{h\to 0} \dfrac{2h^2}{h} = \lim_{h\to 0}(2h) = 0$

3. $\frac{dy}{dx} = \lim_{h\to 0} \frac{8(x+h) - 8x}{h} = \lim_{h\to 0} \frac{8h}{h} = 8$

4. $\dfrac{dy}{dx} = \lim_{h\to 0} \dfrac{3(x+h)^2 - 3x^2}{h} = \lim_{h\to 0} \dfrac{6xh + 3h^2}{h}$

$= \lim_{h\to 0}(6x + 3h) = 6x$

5. $\dfrac{dy}{dx} = \lim\limits_{h\to 0} \dfrac{(x+h)^3 - x^3}{h} = \lim\limits_{h\to 0} \dfrac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h}$

$= \lim\limits_{h\to 0} \dfrac{3x^2h + 3xh^2 + h^3}{h} = \lim\limits_{h\to 0}(3x^2 + 3xh + h^2) = 3x^2$

6. $\dfrac{dy}{dx} = \lim\limits_{h\to 0} \dfrac{(6 - 3(x+h)^2) - (6 - 3x^2)}{h}$

$= \lim\limits_{h\to 0} \dfrac{6 - 3(x^2 + 2xh + h^2) - 6 + 3x^2}{h} = \lim\limits_{h\to 0} \dfrac{-6xh - 3h^2}{h}$

$= \lim\limits_{h\to 0}(-6x - 3h) = -6x$

At $x = 1$, $\frac{dy}{dx} = -6(1) = -6$

12.3 Differentiating x^n

1.
- **a.** $5x^4$
- **b.** $-6x^{-7}$
- **c.** $\frac{1}{2}x^{-1/2}$
- **d.** $-\frac{1}{4}x^{-5/4}$

2.
- **a.** $20x^3$
- **b.** $-10x^{-3}$
- **c.** $-\frac{3}{4}x^{-4}$
- **d.** $-6x^{-5/3}$

3.
- **a.** 54
- **b.** 150
- **c.** 1.5
- **d.** $\frac{27}{8}$

4. $-\frac{15}{16}$

5. $\frac{1}{16}$

6. $\frac{3}{2}$

7. $\frac{8}{9}$

12.4 Differentiating quadratics

1.
- **a.** $6x - 7$
- **b.** $\frac{3}{2}x + \frac{1}{2}$
- **c.** $-10x$

2.
- **a.** -5
- **b.** 24
- **c.** $-\frac{1}{2}$

3. $(3, 13)$

4. -10

5. -2

6.
- **a.** $(2, -1)$ and $(3, 1)$
- **b.** 1 and 3

7.
- **a.** $(-\frac{4}{5}, 7)$ and $(2, 7)$
- **b.** -14 and 14

12.5 Differentiating functions with two or more terms

1.
- **a.** $5 - 14x$
- **b.** $20x^{3/2} - 24x^3$
- **c.** $20x^{3/2} - 8x - 2x^{-2/3}$

2.
- **a.** $\frac{1}{2}x^{-1/2} - \frac{1}{2}x^{-3/2}$
- **b.** $-\frac{2}{7}x^{-3} + \frac{1}{4}x^{-3/4}$

3.
- **a.** $6x^2 - 12x + 5$
- **b.** $3x^2 + \frac{4}{x^2}$
- **c.** $12x - 8$

4.
- **a.** 16
- **b.** -10

5. -3

6. $p = \frac{5}{2}$

7. $p = 9$, $q = -4$

8.
- **a.** $1 + 12x + 60x^2$
- **b.**
  - **i.** $12 + 120x$
  - **ii.** $-12 + 120x$

12.6 Gradients, tangents and normals

1.
- **a.** $y = -4x$
- **b.** $y = 11x + 8$
- **c.** $y = 17x - 28$
- **d.** $y = -\frac{7}{6}x - 4$

2.
- **a.** $y = -\frac{3}{7}x + \frac{109}{14}$
- **b.** $y = \frac{4}{97}x - \frac{4575}{97}$

3.
- **a.** $y = -4x + 11$
- **b.** $y = \frac{1}{4}x + \frac{27}{4}$

4.
- **a.** $y = 5x - 21$
- **b.** $y = -x + 1$
- **c.** $(\frac{11}{3}, -\frac{8}{3})$

5.
- **a.** $y = -x + 2$
- **b.** $(2, 0)$
- **c.** $x = 2 \Rightarrow g(x) = 8 - 4 \times 2 = 0$, so A is on the curve $y = g(x)$

6. $(11, -\frac{43}{2})$

7.
- **a.** $3x - 16y + 28 = 0$
- **b.** $32x + 6y - 143 = 0$
- **c.** $\frac{265}{6}$

12.7 Increasing and decreasing functions

1.
- **a.** $(-1, -\frac{1}{2})$, $(0, 0)$ and $(1, -\frac{1}{2})$
- **b.**
  - **i.** Decreasing
  - **ii.** Increasing
  - **iii.** Decreasing
  - **iv.** Neither
  - **v.** Increasing
  - **vi.** Neither

2.
- **a.** $x \geq \frac{4}{5}$
- **b.** $x \leq -\frac{3}{2}$
- **c.** $x \leq -2$ or $x \geq 2$
- **d.** $-\sqrt{3} \leq x \leq \sqrt{3}$

3.
- **a.** $x \leq -\frac{7}{6}$
- **b.** $x \geq -\frac{3}{5}$
- **c.** All values of x
- **d.** $x \geq \frac{3}{2}$

4. $f'(x) = 5x^4 + 6x^2 + 8$, $5x^4 \geq 0$ and $6x^2 \geq 0$ for all values of x, therefore $5x^4 + 6x^2 + 8 \geq 8 > 0$ for all values of x and therefore $f(x)$ is increasing for all real values of x.

5. $f'(x) = -6x^2 - 36x - 54 = -6(x + 3)^2$
$-6(x + 3)^2 \leq 0$ for all values of x.

6. $q \geq 0$

7.
- **a.** $9x^2 + 4x - 5$
- **b.** $(-\infty, -1] \cup [\frac{5}{9}, \infty)$

12.8 Second order derivatives

1.
- **a.** $18x - 6$, 18
- **b.** $-3x^2 - 4x^{-2}$, $-6x + 8x^{-3}$
- **c.** $-2x^{-3/2} + 12x^{1/2}$, $3x^{-5/2} + 6x^{-1/2}$

2.
- **a.** $-18x^{-4} - 2x^3 - 9$, $72x^{-5} - 6x^2$
- **b.** $-2x^{-5/4} - \frac{1}{6}x^{-5/6}$, $\frac{5}{2}x^{-9/4} + \frac{5}{36}x^{-11/6}$
- **c.** $-5x^4 + 9x^2 + 10x$, $-20x^3 + 18x + 10$

3.
- **a.** $16x^3 - \frac{1}{3}x^{-2/3} - 3x^{-4} - 2x$
- **b.** $48x^2 + \frac{2}{9}x^{-5/3} + 12x^{-5} - 2$

4. $\frac{95}{4}$

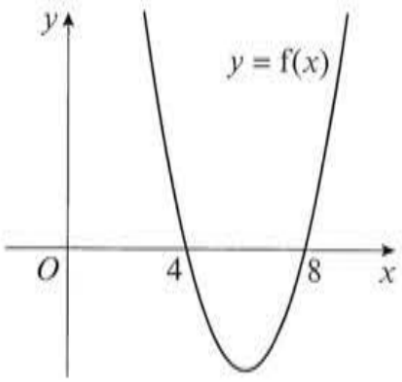
5.
- **a.** $1 - 12x + 48x^2 - 64x^3$
- **b.** $96 - 384x$

6. $a = -3$, $b = 5$

7.
- **a.** $p = -\frac{2}{5}$, $q = \frac{3}{10}$
- **b.** $-\frac{12}{5}x + \frac{3}{5}$

12.9 Stationary points

1. **a.**



b. The curve $y = f(x)$ changes from being a decreasing function to an increasing function between $x = 4$ and $x = 8$, so the curve has a local minimum in this interval.

c. $2x - 12$

d. $x = 6$

e. $(6, -4)$

2.

a. $(3, -1)$

b. $(\frac{3}{2}, -\frac{53}{4})$

c. $(-\frac{1}{3}, \frac{2}{3})$

3.

a. $(-8, 72)$

b. $(\frac{3}{2}, 9)$

c. $(-\frac{15}{7}, \frac{351}{7})$

4.

a. $(-1, -11)$ and $(2, 43)$

b. $-24x + 12$

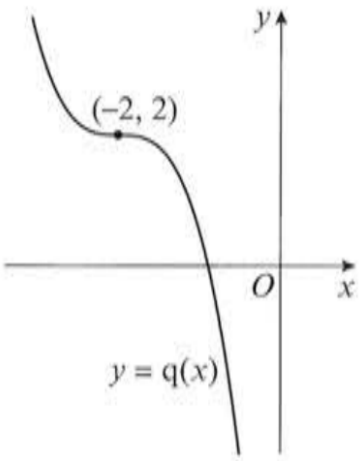
c. $x = -1 \Rightarrow \frac{d^2y}{dx^2} > 0$, so $(-1, -11)$ is a local minimum.
$x = 2 \Rightarrow \frac{d^2y}{dx^2} < 0$, so $(2, 43)$ is a local maximum.

5.

a. $(-2, 2)$

b. $q''(x) = 0$ and the gradient of the curve at $x = -2.1$ and $x = -1.9$ are both negative.

c.



6. $(-\frac{1}{3}, \frac{119}{54})$ and $(\frac{1}{2}, \frac{13}{8})$

7.

a. $(9, 18)$

b. $-x^{-3/2} + 108x^{-3}$

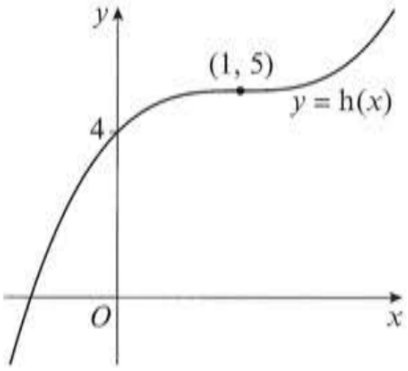
c. $g''(9) = \frac{1}{9} > 0$, so it is a local minimum.

8. $(2, 6)$

9.

a. $(1, 5)$; point of inflection

b.



12.10 Sketching gradient functions

1.

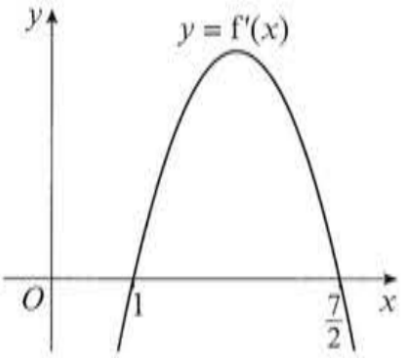
a. $(1, 0)$ and $(\frac{7}{2}, 5)$

b. Positive gradient

c. Negative gradient

d. Negative gradient

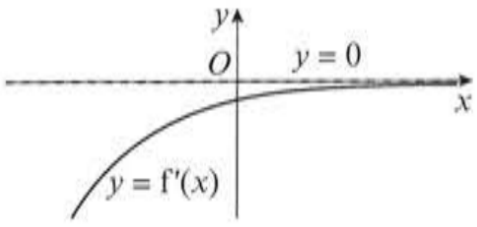
e.



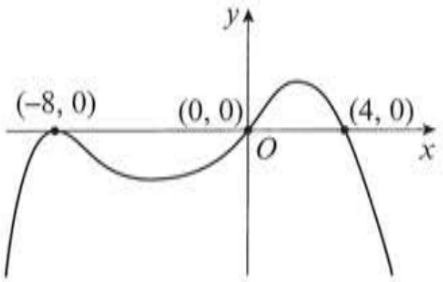
2.

a. The gradient is negative for all values of x.
As x increases, the gradient moves closer to zero.

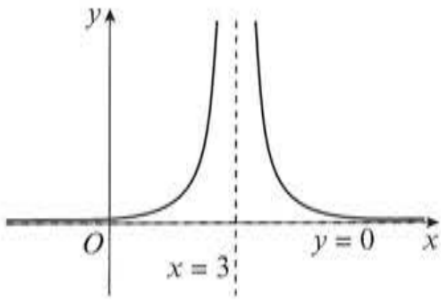
b.



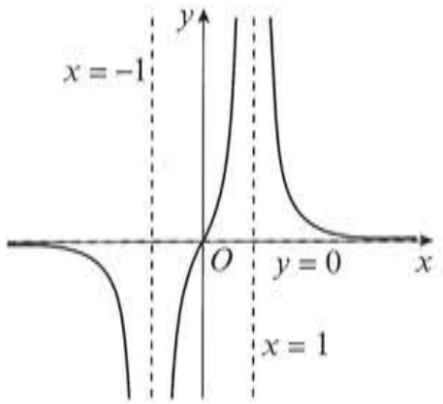
3. **a.**


y
(−8, 0)
(0, 0)
(4, 0)
O
x

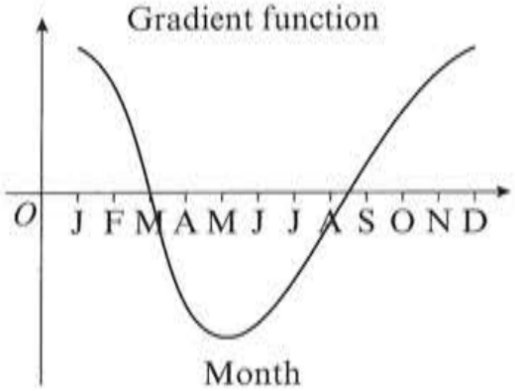
b.


y
O
x = 3
y = 0
x

c.


y
x = −1
O
y = 0
x
x = 1

4.


Gradient function
O
J F M A M J J A S O N D
Month

12.11 Modelling with differentiation

1. $12x$

2.
- **a.** $-40000t^{-3}$
- **b.** -40

3.
- **a.** 64 seconds
- **b.** $-\frac{1}{2}t^{-1/2}$
- **c.** $-\frac{1}{6}$ m s^{-2}

4.
- **a.** $4\pi r^2$
- **b.** 64π cm^3 per cm

5.
- **a.** $y = 40 - x(1+\sqrt{2})$
- **b.** $A = 2xy + x^2 = 2x(40 - x(1+\sqrt{2})) + x^2$
  $= 80x - x^2(1+2\sqrt{2})$
- **c.** $\frac{1600(2\sqrt{2}-1)}{7}$ m^2

6.
- **a.** Substitute $r = \sqrt{\frac{1024}{h}}$ into $S = 2\pi r^2 + 2\pi rh$ and simplify.
- **b.** 384π cm^2

7.
- **a.** Use $2x^2 + 3xy = 608$ and $V = x^2y$ to write V in terms of x.
- **b.** $V = 1360.1$ cm^3
- **c.** Draw a sketch of the graph or find $\frac{d^2V}{dx^2} = -4x$. Interpret sketch or conclude that as $x > 0$, $\frac{d^2V}{dx^2} < 0$, so it is a maximum.

8.
- **a.** $\pi r^2 + 2\pi rh + 2\pi r^2 = 800\pi \Rightarrow h = \frac{800-3r^2}{2r}$

$$V = \pi r^2 h + \frac{2}{3}\pi r^3 = \pi r^2\left(\frac{800-3r^2}{2r}\right) + \frac{2}{3}\pi r^3$$
$$= \frac{\pi r(2400-5r^2)}{6}$$

- **b.** 10600 cm^3

Problem-solving Set A

Bronze

- **a.** $(3, 23)$
- **b.** Local maximum

Silver

- **a.** $(-2, 0)$, $(0, 4)$
- **b.** $x = -2$ and $x = 1$
- **c.** $(-2, 0)$ is a minimum and $(0, 4)$ is a maximum

Gold

- **a.** $(1, -\frac{52}{3})$, $(4, -\frac{56}{3})$
- **b.** $(1, -\frac{52}{3})$ is a maximum and $(4, -\frac{56}{3})$ is a minimum

Problem-solving Set B

Bronze

a. $D(-2,0)$, $E(1,0)$, $F(2,0)$

b. $h'(x) = 3x^2 - 2x - 4$

c.

i. $y = -3x + 3$

ii. $y = 4x - 8$

d. $(\frac{11}{7}, -\frac{12}{7})$

Silver

a. $g'(x) = 3x^2 - 2x - 2$

b.

i. $y = -\frac{1}{3}x - \frac{1}{3}$

ii. $y = -\frac{1}{6}x + \frac{1}{3}$

c. $(-4, 1)$

Gold 9

Selected exam question bank answers

Question 11 $x = \frac{1}{16}$, $x = \frac{81}{16}$

Question 15

a. $f'(x) = \frac{1}{6\sqrt{x}} - \frac{3}{2\sqrt{x^3}}$

b. $(9, 2)$

c. $f''(x) = -\frac{1}{12}x^{-3/2} + \frac{9}{4}x^{-5/2}$
$f''(9) = \frac{1}{162}$ therefore local minimum

Question 27 $a = \frac{8}{9}$, $b = -\frac{5}{3}$, $c = 1$

Question 50

a. $(1, 2.5)$

b. $y = -\frac{2}{7}x + \frac{64}{7}$

c. 128

Question 74

a. Use $P = 300 = 2\pi r + 2y$ to find $y = 150 - \frac{\pi}{2}x$
Substitute into $A = \frac{\pi x^2}{4} + xy$

b. $\frac{22500}{\pi}$ m^2

Question 90

a. Multiply dimensions: x, $36 - 2x$, $60 - 2x$

b. $16 - 2\sqrt{19}$

c. 7092.4 cm^3